

CO5-AT 2 SHORT ANSWER TEST

Name: V Deepa Dharshini

Reg no:192525213

Course Code: CSA0311

Course Name: Data Structures

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CO5-AT2-Short Answer Test [Back](#)

QUESTIONS

- Q1 How many edges must a Minimum Spanning Tree of a graph with V vertices have? 2 marks
- Q2 In a DAG, which node will always have an in-degree of zero in a topological ordering? 2 marks
- Q3 Differences between directed and undirected graphs. 2 marks
- Q4 Define a connected graph. 2 marks
- Q5 What is a topological sort? 2 marks

5/5 answered

Q1 How many edges must a Minimum Spanning Tree of a graph with V vertices have? 2 marks

How many edges must a Minimum Spanning Tree of a graph with V vertices have?

Your Answer

One is the number of edges required for a minimum spanning tree of graph with V vertices.

[Save Answer](#) [Next →](#)

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CO5-AT2-Short Answer Test [Back](#)

QUESTIONS

- Q1 How many edges must a Minimum Spanning Tree of a graph with V vertices have? 2 marks
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- Q3 Differences between directed and undirected graphs. 2 marks
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5/5 answered

Q2 In a DAG, which node will always have an in-degree of zero in a topological ordering? 2 marks

In a DAG, which node will always have an in-degree of zero in a topological ordering?

Your Answer

The source node has an in-degree of zero because it has no in-coming edges.

[Save Answer](#) [Next →](#)

CO5-AT2-Short Answer Test

← Back

QUESTIONS
Q1 How many edges must a Minimum Spanning Tree of a graph with 50 vertices have?
2 marks

Q2 In a DAG, which node will always have an in-degree of zero in a topological ordering?
2 marks

Q3 Differentiate between directed and undirected graphs.
2 marks

Q4 Define a connected graph.
2 marks

Q5 What is a topological sort?
2 marks

3/5 answered

Q3 Differentiate between directed and undirected graphs.

2 marks

Differentiate between directed and undirected graphs.

Your Answer

A directed graph has edges with direction from one vertex to another vertex, while undirected graph has edges without any direction.

Save Answer

Next →

CO5-AT2-Short Answer Test

← Back

QUESTIONS
Q1 How many edges must a Minimum Spanning Tree of a graph with 50 vertices have?
2 marks

Q2 In a DAG, which node will always have an in-degree of zero in a topological ordering?
2 marks

Q3 Differentiate between directed and undirected graphs.
2 marks

Q4 Define a connected graph.
2 marks

Q5 What is a topological sort?
2 marks

5/5 answered

Q4 Define a connected graph.

2 marks

Define a connected graph.

Your Answer

A connect graph is a graph in which there is a path between every pair of vertices

Save Answer

Next →

COS-AT2-Short Answer Test[← Back](#)**QUESTIONS**

Q1 ✓
How many edges must a
Minimum Spanning Tree of
a graph with V vertices
have?
2 marks

Q2 ✓
In a DAG, which node will
always have an in-degree
of zero in a topological
ordering?
2 marks

Q3 ✓
Differentiate between
directed and undirected
graphs.
2 marks

Q4 ✓
Define a connected graph.
2 marks

Q5 ✓
What is a topological sort?
2 marks

5/5 answered

Q1 What is a topological sort?

2 marks

What is a topological sort?

Your Answer:

Topological sort is a linear ordering of vertex such that for every directed edge $u \rightarrow v$, vertex u appears before v .[Save Answer](#)